Multi-Armed Bandit (MAB)

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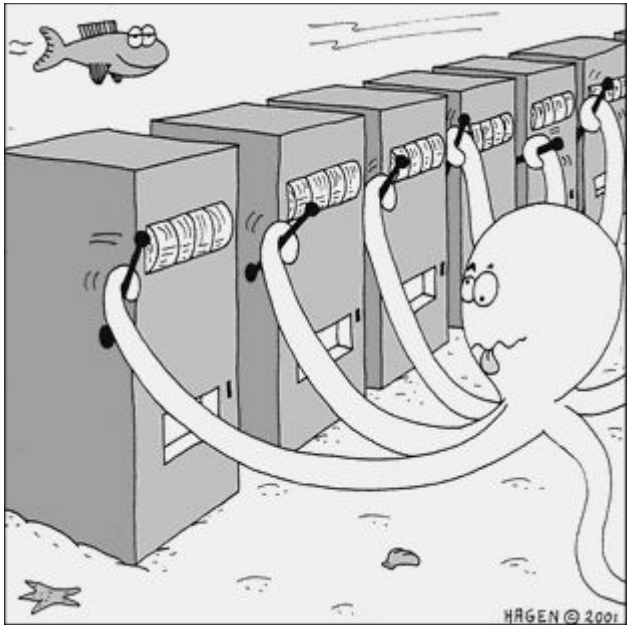
Evaluative feedback vs. Instructive feedback

- What distinguishes reinforcement learning from other types of learning
- The use of training information that evaluates the actions taken rather than *instructs* by giving correct actions
    - **Evaluative feedback** indicates how good the action taken was, but not whether it was the best or the worst action possible
    - **Instructive feedback** indicates the correct action to take, independently of the action actually taken (supervised learning)
- We are going to analyze the fundamental characteristics of evaluative feedback using the Multi-Armed Bandit problem.

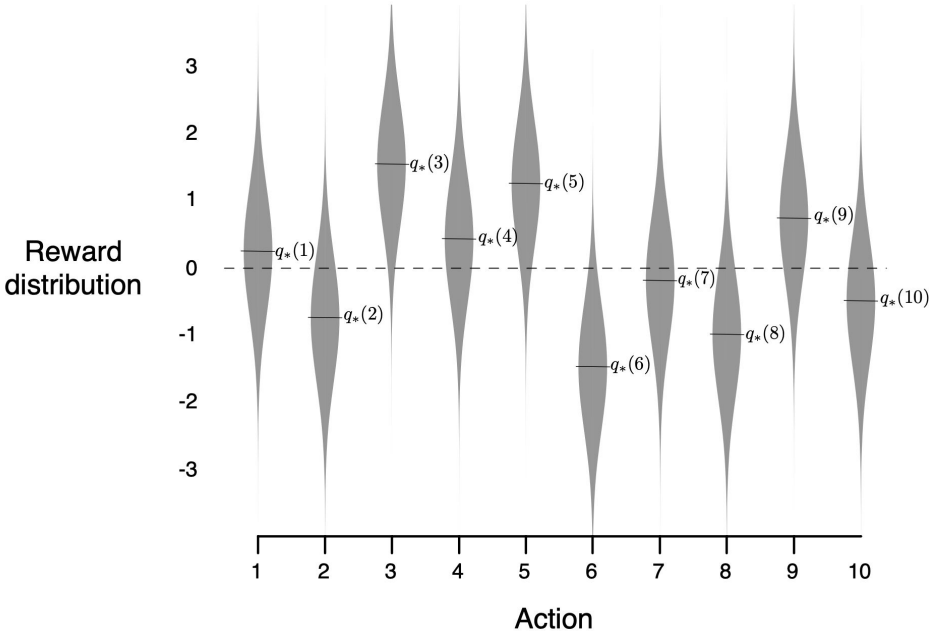
The Bandit

- Assume: k slot machines in front of you
- You will select one of the machines to bet at each round
- After each choice, you receive a numerical reward chosen from a stationary probability distribution
- Goal: find a best slot machine to bet on average



The Bandit

- If you are given the underlying distribution, you win!
- In practice, the distribution is unavailable
- A key is to find out the true expected reward



The Value of an Action

- Similar to the state values, we can define the value of action (slot machine)

$$q_*(a) \doteq \mathbb{E}[R_t \mid A_t = a]$$

- Only way to estimate the value is to collect more rewards
- Exploiting: Repeat good actions based on your current knowledge of the values of the actions
- Exploring: Try new actions or bad actions to improve your estimate of the actions' values
- How do you balance these two?

ϵ-greedy Method

- Choose a random action with probability ϵ
- Use a greedy action with probability $1 - \epsilon$

1	1					100	180	100	110
		2	2	2	4				

Assume the initial estimate is 0

Incremental Implementation

- To compute the average after i-th action, a naive implementation requires a list of all rewards

$$Q_n \doteq \frac{R_1 + R_2 + \cdots + R_{n-1}}{n-1}$$

- The memory and computational requirements would grow over time as more rewards are seen

Incremental Implementation

$$
\begin{aligned}
Q_{n+1} &= \frac{1}{n}\sum_{i=1}^{n} R_i \\
&= \frac{1}{n}\left(R_n + \sum_{i=1}^{n-1} R_i\right) \\
&= \frac{1}{n}\left(R_n + (n-1)\frac{1}{n-1}\sum_{i=1}^{n-1} R_i\right)
\end{aligned}
$$

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&= \frac{1}{n}\left(R_n + (n-1)\frac{1}{n-1}\sum_{i=1}^{n-1} R_i\right) \\
&= \frac{1}{n}\Big(R_n + (n-1)Q_n\Big) \\
&= \frac{1}{n}\Big(R_n + nQ_n - Q_n\Big) \\
&= Q_n + \frac{1}{n}\Big[R_n - Q_n\Big],
\end{aligned}
$$

$$
\boxed{NewEstimate \leftarrow OldEstimate \; + \; StepSize \left[Target - OldEstimate\right]}
$$

ϵ-greedy Method

A simple bandit algorithm

Initialize, for $a = 1$ to k:

$Q(a) \leftarrow 0$

$N(a) \leftarrow 0$

Loop forever:

$$A \leftarrow \begin{cases} \arg\max_a Q(a) & \text{with probability } 1-\varepsilon \quad \text{(breaking ties randomly)} \\ \text{a random action} & \text{with probability } \varepsilon \end{cases}$$

$R \leftarrow bandit(A)$

$N(A) \leftarrow N(A) + 1$

$Q(A) \leftarrow Q(A) + \frac{1}{N(A)}\left[R - Q(A)\right]$

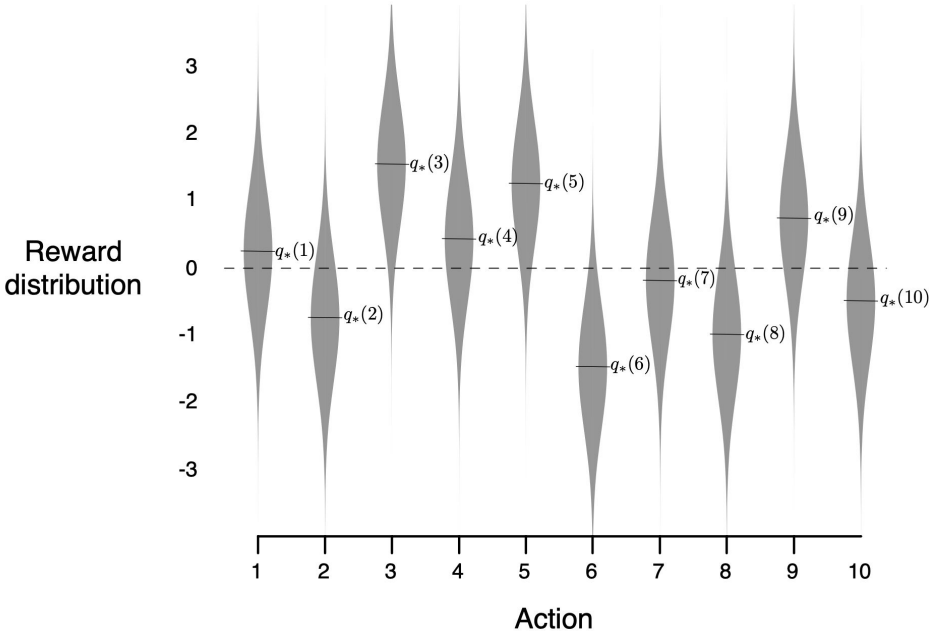
The step size can be defined as a hyper-parameter (needs to be a small value)

Optimistic Initial Values

- Exploration depends to the initial action-value estimates (biased by the initial values)
- Initial action values can also be used as a simple way to encourage exploration
- Where do you want to **explore**? → Slot machines you haven't tried many

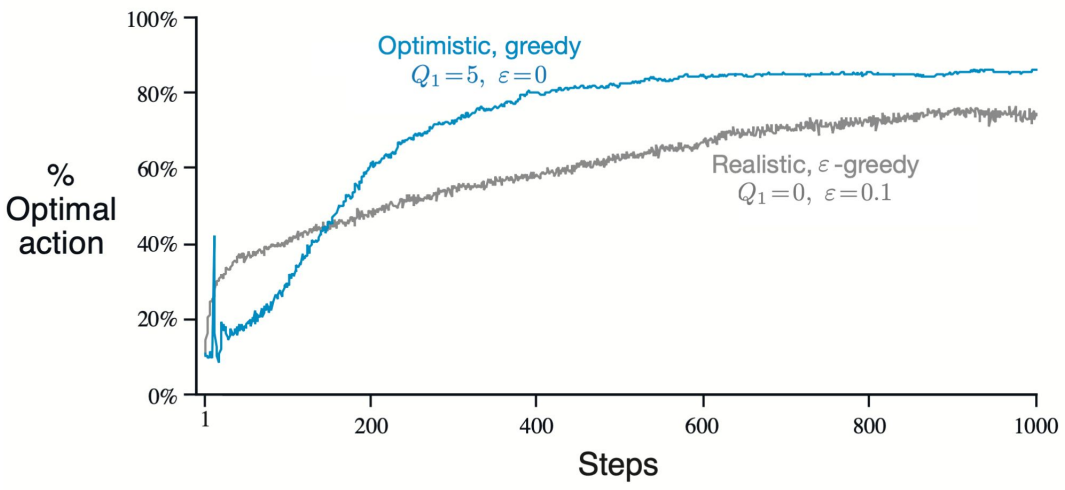
Milestone Questions

- What kind of initialization should we do (to encourage exploration)?
  - Optimistic (much higher than the expected true mean)
  - Realistic (much less than the expected true mean)



Optimistic Initial Values

- e.g. An initial estimate of +5 is wildly optimistic (based on the previous diagram)
- This optimism encourages action-value methods to explore
- The learner switches to other actions, being “disappointed” with the rewards it is receiving



Upper-Confidence-Bound (UCB)

- Is there anyway to quantify the accuracy of the value estimate?
- In a stationary scenario, more samples should give better estimates (Law of Large Numbers)
- We will define the Upper Confidence Bound (UCB) of each arm so that
  - It becomes larger if the arm has not been visited too many times
- Action selection can be greedy based on the current estimate and the UCB

$$a_t = \underset{a \in A}{\operatorname{argmax}} \left(\underline{\overline{Q}_t(a) + U_t(a)} \right)$$

(Estimated Average) + (Uncertainty)

Upper-Confidence-Bound (UCB)

Hoeffding's Inequality

- Let $X_1, X_2, \ldots, X_t$ be i.i.d. random variables in [0, 1], and
- let $\overline{X_t}$ be the sample mean. Then,

$$\mathbb{P}\left[\underline{\mathbb{E}[X]} > \underline{\overline{X_t} + u}\right] \leq e^{-2tu^2}$$

- What is the probability of (the current estimate of average plus some additional value u) exceeding the *true* average E[X]

Upper-Confidence-Bound (UCB)

Hoeffding's Inequality

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- We want to use the Hoeffding's Inequality to guarantee that our greedy based on the UCB rarely exceed the actual Q-values

$$\mathbb{P}\left[Q(a) > \overline{Q_t}(a) + U_t(a)\right] \leq e^{-2N_t(a)U_t(a)^2} (\doteq p)$$

Upper-Confidence-Bound (UCB)

$$p = e^{-2N_t(a)U_t(a)^2}$$

$$U_t(a) = \sqrt{\frac{-\ln p}{N_t(a)}}$$

- Reduce the probability p of the UCB value exceeding the actual Q-values over time (e.g. $p = t^{-4}$)

$$U_t(a) = \sqrt{\frac{2\ln p}{N_t(a)}}$$

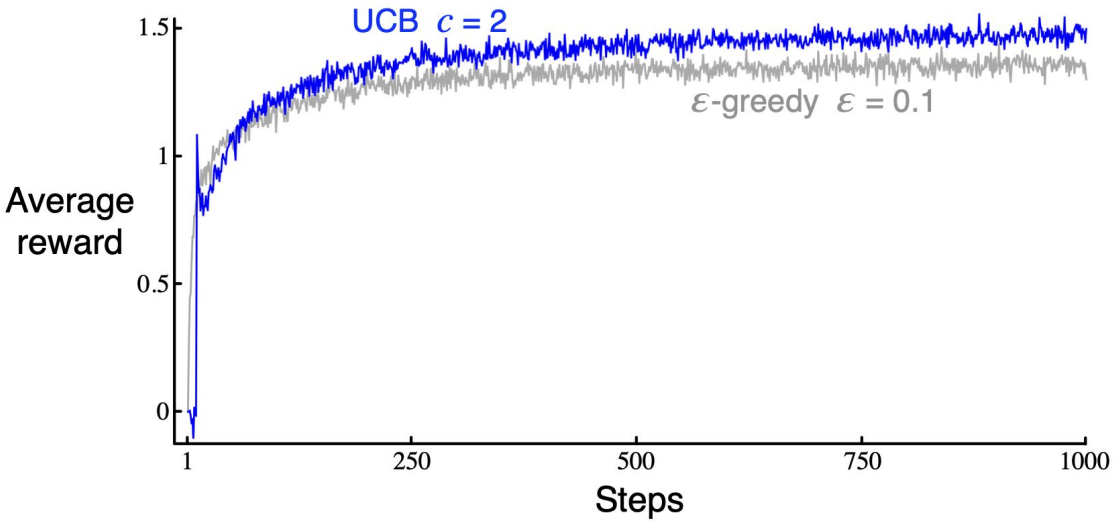
Upper-Confidence-Bound (UCB)

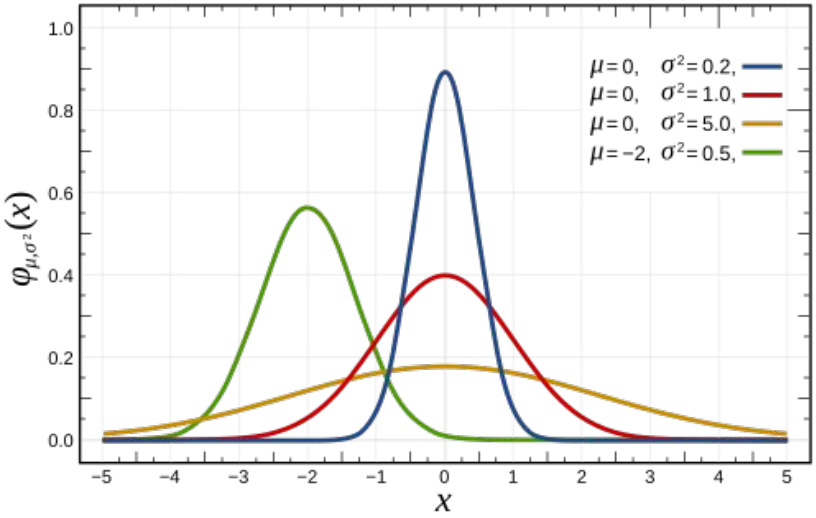
- Quantify the uncertainty about the accuracy of the action-value estimates
- ϵ-greedy action selection forces the non-greedy actions to be tried, but indiscriminately, with no preference for those that are nearly greedy or particularly uncertain

$$a_t = \underset{a \in \mathcal{A}}{\operatorname{argmax}} \left(\bar{Q}(a) + \sqrt{\frac{2 \ln t}{N_t(a)}} \right)$$

Comparing UCB and Epsilon-greedy

- In practice, the epsilon-greedy works really well! (but may require some tuning)
- UCB is a safe choice as it has theoretical guarantee to perform well in general (not necessarily the best)



Demo with Gaussian Bandit

- The epsilon-greedy
- Optimistic initialization
- UCB